

Program 10: Demonstrate k8s dashboard and CLI for clusterMonitoring

Manifest Files:

```
~/Desktop/Home/Rv/devOps/program10 — minikube dashboard ~/D/H/R/d/program10 — kubectl · minikube dashboard  ~/Desktop/Home/Rv/devOps/program10 — ~/D/H/R/d/program10 — fish +

srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program10> cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx

spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx

  template:
    metadata:
      labels:
        app: nginx

    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program10> cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service

spec:
  selector:
    app: nginx

  ports:
    - protocol: TCP
      port: 80
      targetPort: 80

  type: NodePort
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program10> █
```

Deployment and Pods Description

```
srikanths@Srikanths-MacBook-Air ~> minikube start --cpus=2 --memory=2048
🐳 minikube v1.37.0 on Darwin 26.1 (arm64)
🔧 Using the docker driver based on existing profile
⚠️ You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
👉 Starting "minikube" primary control-plane node in "minikube" cluster
🚀 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
  ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, default-storageclass, dashboard
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
srikanths@Srikanths-MacBook-Air ~> cd Desktop/Home/Rv/devOps/
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/devOps> cd program10
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program10> kubectl apply -f service.yaml
service/nginx-service unchanged
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program10> kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment unchanged
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program10> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-sdsqs   1/1     Running   2 (93s ago)    6m7s
nginx-deployment-6f9664446b-zlw6m   1/1     Running   1 (116s ago)   6m7s
srikanths@Srikanths-MacBook-Air ~/D/H/R/d/program10> minikube dashboard
🔍 Verifying dashboard health ...
🚀 Launching proxy ...
🏠 Verifying proxy health ...
🔗 Opening http://127.0.0.1:50373/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
```

The screenshot displays the Kubernetes dashboard interface. The top navigation bar includes the Kubernetes logo, a dropdown menu set to 'default', a search bar, and a notification bell. The left sidebar lists various resource categories: Workloads (Cron Jobs, Daemon Sets, Deployments, Jobs, Pods, Replica Sets, Replication Controllers, Stateful Sets), Service (Ingresses, Ingress Classes, Services), Config and Storage (Config Maps, Persistent Volume Claims, Secrets, Storage Classes), and Cluster (Cluster Role Bindings, Cluster Roles, Events). The main content area is titled 'Workload Status' and features three large green circular indicators representing the status of Deployments, Pods, and Replica Sets, each with a 'Running' count. Below this, the 'Deployments' section shows a table with one entry: 'nginx-deployment' using 'nginx:latest' image, with 2 pods running. The 'Pods' section shows a table with one entry: 'nginx-deployment-6f9664446b-sdsqs' using 'nginx:latest' image, running on the 'minikube' node with 2 restarts.

Name	Images	Labels	Pods	Created
nginx-deployment	nginx:latest	app: nginx	2 / 2	6 minutes ago

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created
nginx-deployment-6f9664446b-sdsqs	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	2	-	-	6 minutes ago